An Overview of the Financial System

Chapter 2

Econ 351 – Monetary Economics

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Preview

- This chapter provides an *overview* of **financial markets** and **institutions**
- How funds flow from savers to borrowers, what instruments are traded, and why the system is regulated

Funds move from savers to borrowers through two channels: **financial markets** (direct finance) and **financial intermediaries** (indirect finance).

Outline

What Is Finance?

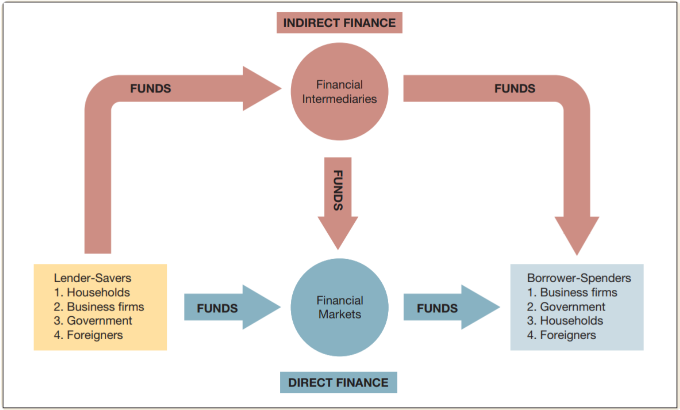
Two Channels of Fund Flows

- **Direct Finance**
  - Funds flow through **financial markets**
  - Savers lend directly via bonds, stocks, etc.
- **Indirect Finance**
  - Funds flow through **financial institutions**
  - Intermediaries collect deposits and make loans

Key Question

Who bears the risk?

Flows of Funds Through the Financial System



Direct finance occurs through financial markets; indirect finance occurs through financial intermediaries.

Source: Mishkin, The Economics of Money, Banking, and Financial Markets

Functions of Financial Markets

Mobilizing Funds

- The most basic and important function
- Financing business investment

Intertemporal Consumption

- Allows households to shift consumption across time
- Example: mortgage lending

Risk Management

- Risk transfer
- Derivatives and asset securitization

Outline

Debt vs. Equity Markets

Market Types

- Debt markets: bonds, loans (creditor claims)
- Equity markets: stocks (ownership claims)

Debt Instruments

- Fixed payments (interest + principal)
- Maturity: short / intermediate / long
- No ownership

Equity Instruments

- Ownership in firm (residual claimants)
- Dividends & voting rights

Market Size

- Debt markets are larger than equity markets (market value)

Debt and Equity: A Spectrum

Pure Debt

- Fixed payments
- Low risk

Pure Equity

- Profit-based return
- High risk

In Between

Preferred stock   Convertible bonds   Performance-based contracts

Debt holders are paid first and get a fixed amount; equity holders are **residual claimants** – they get whatever is left, which is why equity is riskier.

Primary and Secondary Markets

Primary Market

- New securities sold to initial buyers
- Often private transactions
- Investment banks underwrite securities and guarantee prices

Secondary Market

- Previously issued securities
- Examples: NYSE, NASDAQ, bond markets
- Brokers match buyers and sellers; dealers trade directly

The New York Stock Exchange on Wall Street – the best-known secondary market.

Wikimedia Commons, Arild Vågen, CC BY-SA 4.0

Why Are Secondary Markets Important?

Provide Liquidity

- Investors can easily buy and sell securities (e.g., CDs)
- Increases the attractiveness of securities

Price Discovery

- Secondary market prices help determine primary market prices

Improve Market Efficiency

- More accurate valuation of financial instruments
- Reduces price dispersion across markets

Money vs. Capital Markets

Money Market

- Short-term debt instruments
- Maturity: ≤ 1 year
- High liquidity
- Low risk
- Examples: T-bills, commercial paper, CDs

Capital Market

- Long-term debt and equity instruments
- Maturity: > 1 year
- Lower liquidity
- Higher risk
- Examples: stocks, long-term bonds

The dividing line is **maturity**: one year or less is the money market; longer is the capital market.

Outline

Money Market Instruments: Treasury Bills

U.S. Treasury Bills (T-bills)

- Short-term debt issued by the U.S. government
- Maturity: 1, 3, or 6 months
- Sold at a discount (buy low, redeem at face value)
- Most liquid and among the safest securities
- Major holders: banks, households, corporations, intermediaries

A $100,000 U.S. Treasury bill issued in October 1969, due April 1970 – no coupon, sold at a discount.

Wikimedia Commons, JHerbstman, Public domain

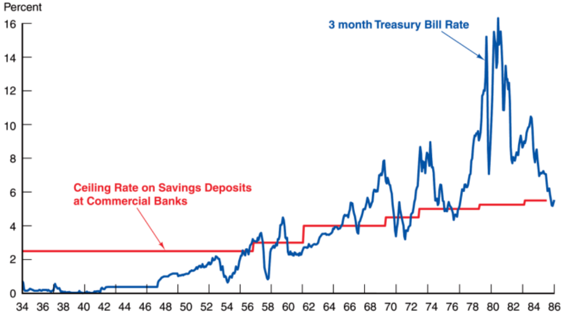
Why Are Money Markets Attractive? (1/2)

Bank Deposits

- Reserve requirements
- Interest rates may be regulated

Money Markets

- Often provide higher yields
- Face fewer regulatory restrictions



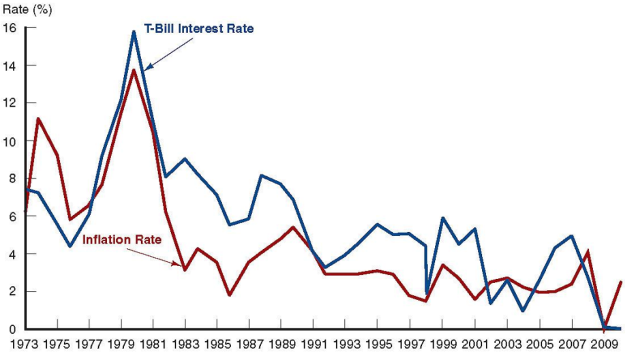
Money market rates versus rates on bank deposits.

Source: Mishkin, The Economics of Money, Banking, and Financial Markets

Why Are Money Markets Attractive? (2/2)

T-bills

- Considered very safe
- Interest rates tend to move with inflation



The T-bill rate and the inflation rate.

Source: Mishkin, The Economics of Money, Banking, and Financial Markets

Money Market Instruments: Negotiable CDs

Negotiable Bank Certificates of Deposit (CDs)

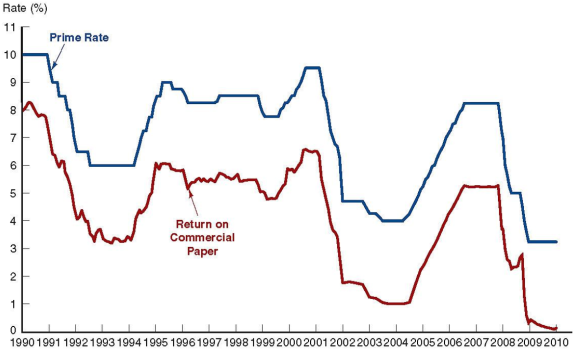
- Bank-issued time deposits: pay interest and return principal at maturity
- Negotiable: can be traded in secondary markets
- High liquidity
- Similar to interest-bearing checking or savings deposits
- Market size (2019): about $1.9 trillion outstanding
- A key funding source for commercial banks

> Because a negotiable CD can be resold before maturity, the depositor is not locked in – that liquidity is what makes it a money market instrument.

Money Market Instruments: Commercial Paper

Commercial Paper

- Short-term unsecured debt by large banks and corporations
- Provides a quick and low-cost source of funding



Commercial paper rate and the T-bill rate.

Source: Mishkin, The Economics of Money, Banking, and Financial Markets

Money Market Instruments: Repos

Repurchase Agreements (Repos)

- Short-term collateralized loans (often overnight to two weeks)
- Collateral typically: Treasury bills or other government securities
- Economically similar to secured (collateralized) lending
- Important source of short-term funding for banks
- Major lenders: large corporations and financial institutions

> Microsoft lends $1 million by purchasing Treasury bills from a bank, which repurchases them at a slightly higher price the next week.

Capital Market Instruments

Capital Market Instruments

- Long-term debt and equity instruments (maturity > 1 year)
- Larger price fluctuations than money market instruments
- Generally riskier investments
- Examples:
  - Stocks
  - Corporate bonds
  - Government and agency securities
  - Municipal bonds
  - Mortgages and mortgage-backed securities
  - Consumer and commercial loans

Capital Market Instruments: Municipal Bonds

State and Local Government (Municipal) Bonds

- Issued by state and local governments
- Finance schools, roads, and public projects
- Interest is exempt from federal (and often state) income taxes
- Especially attractive to **high marginal tax rate** investors
- Major holders: commercial banks, wealthy individuals, insurance companies

A tax-exempt 4% municipal bond is worth more to an investor in the 37% bracket than to one in the 10% bracket – the higher your marginal rate, the more the exemption is worth.

U.S. Federal Income Tax: Marginal Rates

Marginal Rate	Single Filers	Married Filing Jointly
10%	\$0 – \$11,000	\$0 – \$22,000
12%	\$11,001 – \$44,725	\$22,001 – \$89,450
22%	\$44,726 – \$95,375	\$89,451 – \$190,750
24%	\$95,376 – \$182,100	\$190,751 – \$364,200
32%	\$182,101 – \$231,250	\$364,201 – \$462,500
35%	\$231,251 – \$578,125	\$462,501 – \$693,750
37%	Over \$578,125	Over \$693,750

- Higher income is subject to higher marginal tax rates
- High-income individuals have stronger incentives to engage in tax planning

Capital Market Instruments: Mortgages and MBS

Mortgages

- Loans for land, housing, or structures, backed by collateral
- Residential mortgages dominate the market

Mortgage-Backed Securities (MBS)

- Bond-like securities backed by pools of mortgages
- Played an important role in the 2007–2009 financial crisis

Key Federal Agencies

- Fannie Mae, Freddie Mac, Ginnie Mae
- Support mortgage markets by purchasing mortgages and issuing securities

Capital Market Instruments: Government and Agency Securities

U.S. Government Securities

- Issued by the U.S. Treasury to finance federal deficits
- Most liquid capital market securities
- Major holders: Federal Reserve, banks, households, foreign investors

Agency Securities

- Issued by government-sponsored or federal agencies
- Used to support mortgages, farm loans, and infrastructure projects
- Often (explicitly or implicitly) guaranteed by the federal government

Outline

International Bond Market

Foreign Bonds

- Issued by country A in country B
- Denominated in country B's currency
- Example: German firm issues a U.S. dollar bond in the United States

Eurobonds

- Issued by country A in country B
- Denominated in country C's currency
- Example: U.S. dollar bond issued in London

The test is the **currency**: a foreign bond is in the currency of the country where it is sold; a Eurobond is in some other currency.

Eurocurrencies and Eurodollars

Eurocurrencies

- Foreign currencies deposited outside their home countries
- Example: euros deposited in a U.S. bank

Eurodollars

- U.S. dollars deposited outside the United States
- Important source of short-term funds for banks

Clarification

- “Euro” does **not** refer to the euro currency
- Many Eurobonds are denominated in U.S. dollars

Outline

Functions of Financial Intermediaries

Lower Transaction Costs

- Economies of scale
- Liquidity services

Reduce Risk

- Risk sharing (asset transformation)
- Diversification

Address Asymmetric Information

- Adverse selection (before the transaction)
- Moral hazard (after the transaction)

Asymmetric Information Problems

Adverse Selection (Before the Transaction)

- Risky borrowers are more likely to seek loans
- Example: risky vs. safe borrower
- Lenders may avoid lending without good information

Moral Hazard (After the Transaction)

- Borrowers may take hidden actions that increase risk
- Example: using loan for unintended purposes
- Lenders may suffer higher losses

How Intermediaries Help

- Screening $\rightarrow$ reduces adverse selection
- Monitoring $\rightarrow$ reduces moral hazard

Outline

Depository Institutions

Commercial Banks

- Liabilities: checkable, savings, and time deposits
- Assets: commercial, consumer, and mortgage loans; government and municipal bonds

Savings Institutions (S&Ls, Mutual Savings Banks)

- Funded by savings, time, and checkable deposits
- Originally focused on residential mortgages; now more diversified

Credit Unions

- Small cooperative institutions serving specific groups
- Primary assets: consumer loans

Contractual Savings Institutions

Life Insurance Companies

- Funded by insurance premiums
- Invest mainly in corporate bonds and mortgages
- Limited holdings of stocks

Fire and Casualty Insurance Companies

- Funded by insurance premiums
- Invest heavily in liquid assets (e.g., municipal bonds)

Pension and Government Retirement Funds

- Funded by employer and employee contributions
- Invest mainly in corporate bonds and stocks

Investment Intermediaries

Finance Companies

- Funded by issuing commercial paper, stocks, and bonds
- Lend to consumers and businesses

Mutual Funds

- Pool funds to buy diversified portfolios of stocks and bonds
- Lower transaction costs and provide diversification

Money Market Mutual Funds

- Invest in money market instruments
- Offer check-writing features similar to deposit accounts

Hedge Funds

- Limited partnerships with high minimum investments
- Use diverse or specialized investment strategies

Outline

Why Regulate the Financial System?

Increase Information for Investors

- Reduce adverse selection and moral hazard
- Improve market efficiency

Ensure Soundness of Financial Intermediaries

- Prevent financial panics
- Protect economic stability

Both goals trace back to **asymmetric information**: investors cannot easily tell good securities (or sound institutions) from bad ones, so regulation forces disclosure and limits risk-taking.

SEC and Investor Protection

Key Laws and Institutions

- Securities Act of 1933
- Securities and Exchange Commission (SEC), 1934

Disclosure Requirements

- Sales, assets, and earnings
- Restrictions on insider trading

Impact

- Greater transparency
- Higher investor confidence

Ensuring Soundness of Financial Intermediaries

Restrictions on Entry

- Charters required
- Minimum capital and credibility standards

Prudential Regulation

- Disclosure and reporting
- Restrictions on assets and activities

Safety Nets

- Deposit insurance (FDIC up to $250,000)

Other Regulations

- Limits on competition (historically)
- Restrictions on interest rates (e.g., Regulation Q)

Financial Regulation Abroad

Similarities Across Countries

- Licensing and supervision
- Disclosure requirements
- Prohibition of insider trading

Key Differences

- Branching restrictions: the U.S. historically restricted bank branches (abolished in 1994)
- Asset holdings: U.S. banks face stricter limits on holding commercial shares than banks in Japan and Germany

Chapter Summary

- Funds flow from savers to borrowers directly through financial markets or indirectly through intermediaries
- Markets are classified by claim (debt vs. equity), by issue (primary vs. secondary), and by maturity (money vs. capital market)
- Money market instruments (T-bills, CDs, commercial paper, repos) are short-term, liquid, and safe; capital market instruments (stocks, bonds, mortgages) are long-term and riskier
- Intermediaries lower transaction costs, share risk, and reduce adverse selection and moral hazard through screening and monitoring
- Regulation increases information for investors and ensures the soundness of intermediaries